

Richard Wu

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EDUCATION

University at Buffalo | Buffalo, NY

- Bachelor of Science, Computer Science
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WORK EXPERIENCE

New Design High School

July 2023 - Present

Coding and Robotics Instructor/Technology Support Specialist

- Designed and delivered project-based robotics curriculum teaching core programming concepts - including loops, conditionals, event-driven logic - using Sphero BOLT platforms
 - Applied Agile-inspired workflows to mentor students through systematic debugging and iterative development cycles, reinforcing real-world software practices
 - Eliminated 13+ hours of manual work per cycle by engineering a Google Apps Script automation that consolidated 10+ spreadsheets into 400+ individualized progress reports with conditional formatting, custom templates, and automated email distribution
 - Deployed automated absence tracking system syncing staff schedules to centralized calendars, eliminating manual data entry for 50+ staff members
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PROJECTS

ARQS: [React](#), [TypeScript](#), [Node.js](#), [Express](#), [Postgresql](#), [Docker](#), [Railway](#) - [GitHub](#)

Jan 2026 - Present

Community Character Glossary [Full-Stack Developer]

- Built a full-stack Wikipedia-style platform allowing for community contributions and real-time glossary management, deployed on Railway with CI/CD
- Designed and built an intuitive UI that streamlines the full content lifecycle—browsing, editing, and resolving conflicting changes—within a single cohesive workflow
- Integrated the OpenLibrary API to dynamically enrich pages with book data, including covers, author details, and publication metadata
- Eliminated a key performance bottleneck by refactoring algorithm to consolidate redundant API calls into single call at initialization, reducing call volume by roughly 5x and improving response times

numBRR: [React](#), [TypeScript](#), [Node.js](#), [Render](#), [Socket.IO](#), [Sass](#) - [GitHub](#)

Dec 2024 - April 2025

Browser-Based Multiplayer Auto-Battler Game [Full-Stack Developer]

- Reduced sync latency by designing a real-time event-driven architecture with Socket.IO and streamlined React state management via custom hooks and Context API in TypeScript
- Implemented scalable matchmaking on Node.js/Socket.IO (Render) using room partitioning and targeted event broadcasting to support 200+ concurrent users
- Formulated a template-driven item system using weighted randomness and parametric inheritance, reducing new feature prototyping time by 60% while driving a 30% improvement in user engagement via level-based drop rates
- Optimized UI with React hooks and prioritized visual hierarchy, cutting first-time user errors by 50%

Collaj: [MongoDB](#), [Express](#), [React](#), [Node.js](#), [Render](#), [JavaScript](#) - [GitHub](#)

Sep 2022 - Nov 2022

MERN Dynamic Art Gallery (Hue-Sorted Entries) [Full-Stack Developer]

- Implemented RESTful API with Express/Node.js and MongoDB for high-performance CRUD operations
 - Engineered Puppeteer scraper to automate extraction of artwork metadata from dynamic web pages
 - Secured user sessions via JWT (JSON Web Tokens)/BCrypt authentication and encrypted data storage
 - Validated API reliability with Mocha/Chai test suites for performance and edge-case handling
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TECHNICAL SKILLS

Skills: React, Python, JavaScript, TypeScript, Node.js, HTML + CSS, Postgresql, Docker, Figma, Git